CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 6: Sets

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write in roster form: $\{x : x \text{ is a prime number less than } 20\}$.
2. Write in set-builder form: $\{5, 10, 15, 20, 25\}$.
3. State whether each is finite, infinite or empty: (i) multiples of 7 (ii) months with 32 days (iii) factors of 24.
4. Are $\{2, 4, 6\}$ and $\{a, b, c\}$ equal or equivalent?
5. How many subsets does $\{p, q, r, s\}$ have?
6. If $A = \{1, 3, 5, 7\}$ and $B = \{3, 4, 5, 6\}$, find $A \cup B$ and $A \cap B$.
7. For the sets above, find $A - B$ and $B - A$.
8. If $\xi = \{1, 2, \ldots, 9\}$ and $A = \{2, 4, 6, 8\}$, find A'.
9. $n(A) = 15$, $n(B) = 12$ and $n(A \cup B) = 20$. Find $n(A \cap B)$.
10. Of 50 students, 30 like tea, 25 like coffee and 10 like both. How many like neither?

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Answer Key

1. $\{2, 3, 5, 7, 11, 13, 17, 19\}$
2. $\{x : x = 5n, n \in N, n \leq 5\}$
3. (i) Infinite (ii) Empty (iii) Finite
4. Equivalent, not equal
5. 16
6. $\{1, 3, 4, 5, 6, 7\}$ and $\{3, 5\}$
7. $\{1, 7\}$ and $\{4, 6\}$
8. $\{1, 3, 5, 7, 9\}$
9. 7
10. 5

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